

Data Analytics — Power BI Project - 1

Data Transformation & Data Modeling

Prepared By: Dhanalakshmi A

Date: 18/06/2026

This document walks through every Power Query transformation and data-modeling step performed for this assignment on the three source files: **List of Orders.csv**, **Order Details.csv**, and **Sales Target.csv**. Each section explains the exact Power BI step to perform, why it is needed, and shows the actual result produced by applying that logic to the real dataset.

As a Data Analyst, this assignment prepares and models an e-commerce sales dataset in Power BI. The raw data is split across three related files: order-level information, product/line-item details, and monthly category sales targets. The goal is to clean, transform, merge, and relate these tables so they can support reliable profit and target-performance analysis.

Bringing in Data and Cleaning It (Power Query)

1.Data Import

I brought three data files into Power BI using the **Get Data** button. Then, I clicked **Transform Data** to open the Power Query Editor so I could clean the tables.

- **Tables Used:** List of Orders, Order Details, and Sales Target.

2.Keeping 500 Rows

The List of Orders table had too many rows. To keep the project simple, I only kept the first 500 rows.

- **How I did it:** I went to Home ----> Keep Rows ----> Keep Top Rows and typed in 500.
- **Why:** To work with a smaller, cleaner set of data.

3.Changing Data Types

I checked the columns and changed their data types so Power BI can do math correctly.

- Changed Order Date to **Date**.
- Changed Amount to **Fixed Decimal Number**
- Changed Target to **Fixed Decimal Number**
- **Why:** Correct data types make the report run faster and prevent mistakes.

4.Fixing Text Formatting

The names in the CustomerName column were messy with random capital letters. I fixed them so they look neat.

- **How I did it:** I clicked Customer Name, then went to Transform -----> Format -----> Capitalize Each Word.

Data Analytics — Power BI Project - 1

5. Making a Location Column

I combined the City column and the State column into one new column called Location.

- **The Rule:** Location = [City] & ", " & [State]
- **How I did it:** I selected both columns, then went to Transform -----> Merge Columns. I chose a *Comma* as the separator and named the new column Location.
- **Why:** This makes it easier to show cities and states on a map visual.

Custom Column

Add a column that is computed from the other columns.

New column name

Location

Custom column formula ⓘ

= [City]&" , " & [State]

Available columns

Order ID
Order Date
CustomerName
State
City

<< Insert

[Learn about Power Query formulas](#)

✓ No syntax errors have been detected.

OK

Cancel

Creating New Columns and Combining Tables

6. Finding the Profit Margin

I made a new column to calculate the percentage of profit for each sale.

- **The Math:**

{Profit Margin} = Profit/Amount

- **How I did it:** I went to Add Column -----> Custom Column, typed the formula, and changed the format to a percentage (%).
- **Why:** It shows how profitable the sales are, not just how much money came in.

Data Analytics — Power BI Project - 1

Custom Column

Add a column that is computed from the other columns.

New column name

Custom column formula ①

= [Profit]/[Amount]

Available columns

- Order ID
- Amount
- Profit
- Quantity
- Category
- Sub-Category

<< Insert

[Learn about Power Query formulas](#)

✓ No syntax errors have been detected.

OK Cancel

7.Labeling Profit Status

I created a special column called Profit Status to instantly tell me if an order made money or lost money.

Add Conditional Column

Add a conditional column that is computed from the other columns or values.

New column name

	Column Name	Operator	Value ①		Output ②
If	Profit Margin	is greater than	ABC 123 90	Then	ABC 123 PROFIT
Else If	Profit Margin	equals	ABC 123 50	Then	ABC 123 BREAK EVEN ...

Add Clause

Else ①

ABC 123

LOSS

OK Cancel

- **The Rules:**
 - If Profit is less than 0 -----> **"Loss"**
 - If Profit equals 0 -----> **"Break-Even"**
 - If Profit is more than 0 -----> **"Profit"**

Data Analytics — Power BI Project - 1

Queries [3]
List of Orders
Order Details
Sales target

fx = Table.AddColumn(#"Profit Margin data type(%)", "Profit Status", each if [Profit Margin] > 0 then "Profit" else "Loss")

	Quantity	Category	Sub-Category	% Profit Margin	Profit Status
1	-1,148.00	7 Furniture	Bookcases	-90.04%	Loss
2	-12.00	5 Clothing	Stole	-18.18%	Loss
3	-2.00	3 Clothing	Hankerchief	-25.00%	Loss
4	-56.00	4 Electronics	Electronic Games	-70.00%	Loss
5	-111.00	2 Electronics	Phones	-66.07%	Loss
6	-272.00	5 Electronics	Phones	-64.15%	Loss
7	1,151.00	4 Electronics	Phones	43.98%	Profit
8	212.00	3 Clothing	Saree	37.79%	Profit
9	-5.00	8 Clothing	Saree	-4.20%	Loss
10	-60.00	5 Clothing	Trousers	-4.43%	Loss
11	-30.00	1 Furniture	Chairs	-125.00%	Loss
12	-166.00	3 Clothing	Saree	-86.01%	Loss
13	5.00	3 Clothing	Trousers	2.78%	Profit
14	16.00	4 Clothing	Stole	13.79%	Profit
15	36.00	6 Clothing	Stole	33.64%	Profit
16	1.00	2 Clothing	Hankerchief	8.33%	Profit
17	18.00	1 Clothing	Kurti	47.37%	Profit
18	17.00	2 Clothing	T-Shirt	26.15%	Profit
19	5.00	9 Clothing	Saree	3.18%	Profit
20	0.00	7 Furniture	Stole	0.00%	Loss

8 COLUMNS, 999+ ROWS Column profiling based on top 1000 rows

PREVIEW DOWNLOADED AT 01:55 PM

- **Why:** This helps us easily filter good orders from bad ones on the charts.

8.Merging Tables Together

I combined the List of Orders table and the Order Details table into one big table using a shared ID.

- **How I did it:** I went to Home -----> Merge Queries as New. I selected both tables, clicked on Order ID in both, and used a **Left Outer Join**.
- **New Table Name:** Orders Data
- **Why:** This puts all the order information and product details into one easy-to-use table.

9.Checking for Missing Data

I used a tool in Power BI to see if there were any blank or empty spaces in the data.

- **How I did it:** I went to View -----> checked Column Quality.
- **Result:** Because I limited the data to 500 rows, there were zero missing values. I did not need to delete or fix anything.

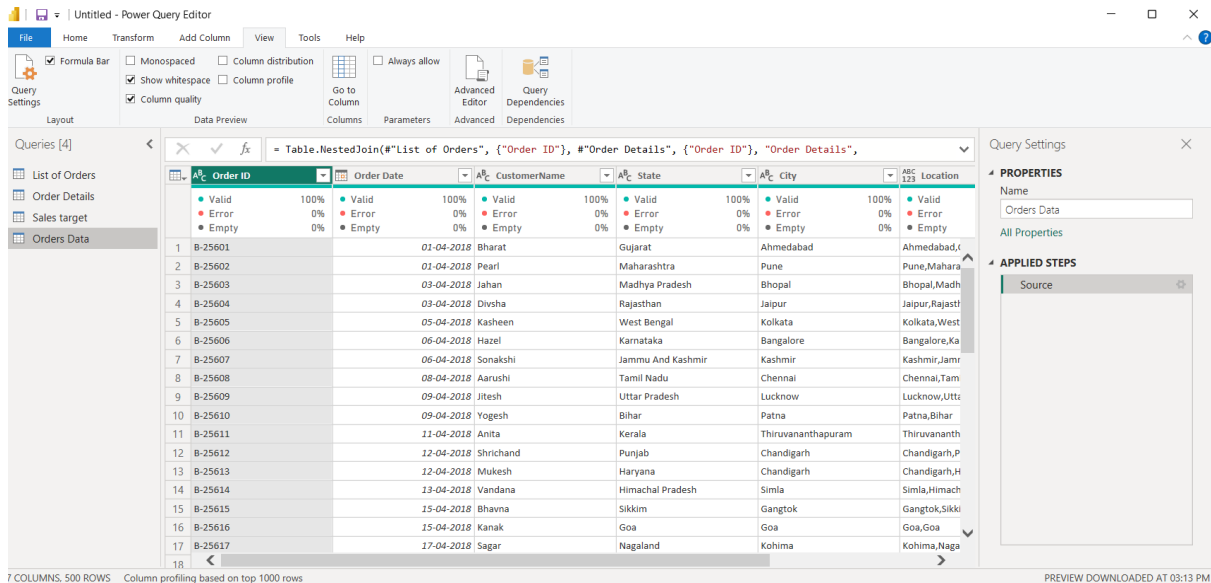
Checking for Double Entries (Duplicates)

I checked to make sure no orders were accidentally copied twice.

- **How I did it:** I went to Home -----> Remove Rows -----> Remove Duplicates.

Data Analytics — Power BI Project - 1

- **Result:** The data was already completely unique. No rows were removed.



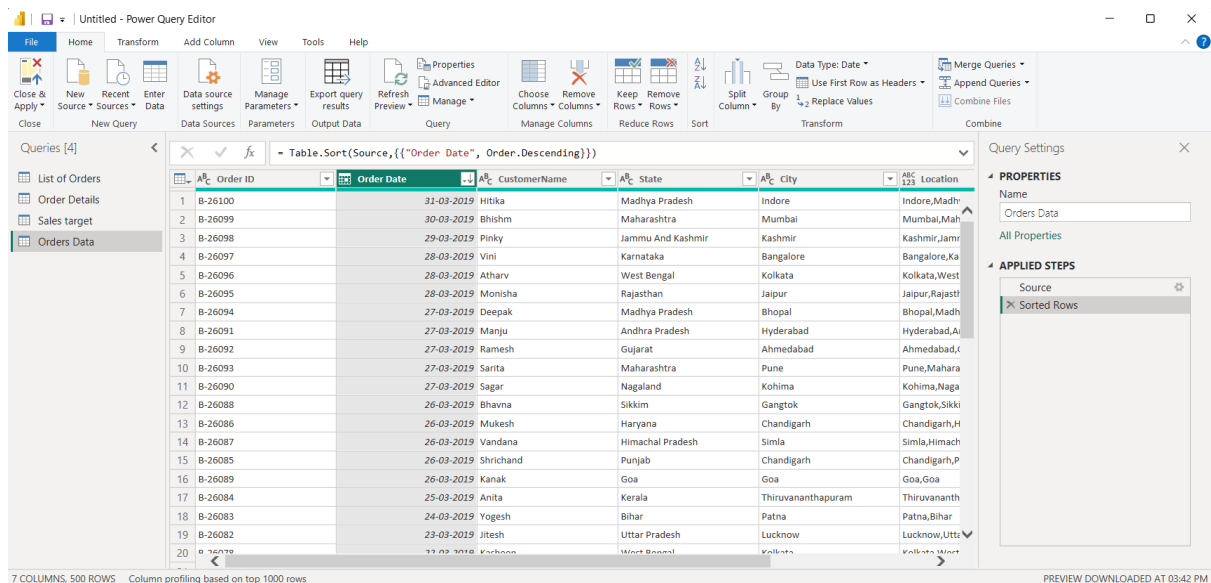
Power Query Editor interface showing a table of order data. The table has columns: Order ID, Order Date, CustomerName, State, City, and Location. The data is sorted by Order Date in descending order.

Order ID	Order Date	CustomerName	State	City	Location
B-25601	01-04-2018	Bharat	Gujarat	Ahmedabad	Ahmedabad, Gujarat
B-25602	01-04-2018	Pearl	Maharashtra	Pune	Pune, Maharashtra
B-25603	03-04-2018	Jahan	Madhya Pradesh	Bhopal	Bhopal, Madhya Pradesh
B-25604	03-04-2018	Divsha	Rajasthan	Jaipur	Jaipur, Rajasthan
B-25605	05-04-2018	Kasheen	West Bengal	Kolkata	Kolkata, West Bengal
B-25606	06-04-2018	Hazel	Karnataka	Bangalore	Bangalore, Karnataka
B-25607	06-04-2018	Sonakshi	Jammu And Kashmir	Kashmir	Kashmir, Jammu And Kashmir
B-25608	08-04-2018	Aarushi	Tamil Nadu	Chennai	Chennai, Tamil Nadu
B-25609	09-04-2018	Jitesh	Uttar Pradesh	Lucknow	Lucknow, Uttar Pradesh
B-25610	09-04-2018	Yogesh	Bihar	Patna	Patna, Bihar
B-25611	11-04-2018	Anita	Kerala	Thiruvananthapuram	Thiruvananthapuram, Kerala
B-25612	12-04-2018	Shrichand	Punjab	Chandigarh	Chandigarh, Punjab
B-25613	12-04-2018	Mukesh	Haryana	Chandigarh	Chandigarh, Haryana
B-25614	13-04-2018	Vandana	Himachal Pradesh	Simla	Simla, Himachal Pradesh
B-25615	15-04-2018	Bhavna	Sikkim	Gangtok	Gangtok, Sikkim
B-25616	15-04-2018	Kanak	Goa	Goa	Goa, Goa
B-25617	17-04-2018	Sagar	Nagaland	Kohima	Kohima, Nagaland

Sorting the Dates

I sorted the data so the newest orders show up first.

- **How I did it:** I clicked the arrow next to Order Date and chose **Sort Descending**.
- **Why:** This allows the user to see the most recent sales immediately.



Power Query Editor interface showing a table of order data. The table has columns: Order ID, Order Date, CustomerName, State, City, and Location. The data is sorted by Order Date in descending order.

Order ID	Order Date	CustomerName	State	City	Location
B-26100	31-03-2019	Hitika	Madhya Pradesh	Indore	Indore, Madhya Pradesh
B-26099	30-03-2019	Bhishm	Maharashtra	Mumbai	Mumbai, Maharashtra
B-26098	29-03-2019	Pinky	Jammu And Kashmir	Kashmir	Kashmir, Jammu And Kashmir
B-26097	28-03-2019	Vini	Karnataka	Bangalore	Bangalore, Karnataka
B-26096	28-03-2019	Atharv	West Bengal	Kolkata	Kolkata, West Bengal
B-26095	28-03-2019	Monisha	Rajasthan	Jaipur	Jaipur, Rajasthan
B-26094	27-03-2019	Deepak	Madhya Pradesh	Bhopal	Bhopal, Madhya Pradesh
B-26091	27-03-2019	Manju	Andhra Pradesh	Hyderabad	Hyderabad, Andhra Pradesh
B-26092	27-03-2019	Ramesh	Gujarat	Ahmedabad	Ahmedabad, Gujarat
B-26093	27-03-2019	Sarita	Maharashtra	Pune	Pune, Maharashtra
B-26090	27-03-2019	Sagar	Nagaland	Kohima	Kohima, Nagaland
B-26088	26-03-2019	Bhavna	Sikkim	Gangtok	Gangtok, Sikkim
B-26086	26-03-2019	Mukesh	Haryana	Chandigarh	Chandigarh, Haryana
B-26087	26-03-2019	Vandana	Himachal Pradesh	Simla	Simla, Himachal Pradesh
B-26085	26-03-2019	Shrichand	Punjab	Chandigarh	Chandigarh, Punjab
B-26089	26-03-2019	Kanak	Goa	Goa	Goa, Goa
B-26084	25-03-2019	Anita	Kerala	Thiruvananthapuram	Thiruvananthapuram, Kerala
B-26083	24-03-2019	Yogesh	Bihar	Patna	Patna, Bihar
B-26082	23-03-2019	Jitesh	Uttar Pradesh	Lucknow	Lucknow, Uttar Pradesh
B-26079	22-03-2019	Karishma	West Bengal	Kolkata	Kolkata, West Bengal

Filtering for One State

I set up a filter to look at data from just one specific area.

- **How I did it:** I went to the Filter options, selected State, and checked the box for **Tamil Nadu**.
- **Why:** To practice analyzing sales for a single region.

Data Analytics — Power BI Project - 1

The screenshot shows the Power Query Editor interface. The main area displays a table with 8 columns and 8 rows of data. The columns are: Order ID, Order Date, CustomerName, State, City, and Location. The data is filtered for the state of Tamil Nadu. The right-hand pane shows the 'Query Settings' for the 'Filtered Rows' step.

Order ID	Order Date	CustomerName	State	City	Location
B-26081	22-03-2019	Aarushi	Tamil Nadu	Chennai	Chennai,Tamil Ni
B-26018	14-02-2019	Aarushi	Tamil Nadu	Chennai	Chennai,Tamil Ni
B-26008	09-02-2019	Kalyani	Tamil Nadu	Chennai	Chennai,Tamil Ni
B-25860	15-11-2018	Akshay	Tamil Nadu	Chennai	Chennai,Tamil Ni
B-25788	21-09-2018	Dinesh	Tamil Nadu	Chennai	Chennai,Tamil Ni
B-25716	11-07-2018	Surabhi	Tamil Nadu	Chennai	Chennai,Tamil Ni
B-25698	23-06-2018	Amisha	Tamil Nadu	Chennai	Chennai,Tamil Ni
B-25608	08-04-2018	Aarushi	Tamil Nadu	Chennai	Chennai,Tamil Ni

Grouping Data

I made copies of my tables and grouped them to find total numbers for different categories.

Counting Total Orders

- *What I did:* I copied Order Details, went to Group By, and chose **Count Rows**. I named it Order Count.
- *Why:* To see exactly how many orders were placed.

Finding Average Profit by Category

- *What I did:* I grouped the data by Category and calculated the **Average** of the Profit column.
- *Why:* To see which product types make the most money on average.

The screenshot shows the Power Query Editor interface. The main area displays a table with 2 columns: Category and Category Profit. The data is grouped by Category. The right-hand pane shows the 'Query Settings' for the 'Grouped Rows' step.

Category	Category Profit
Furniture	9.456790123
Clothing	11.76290832
Electronics	34.07142857

Finding Total Sales by Sub-Category

Data Analytics — Power BI Project - 1

- *What I did:* I grouped the data by Sub-Category and calculated the **Sum** (total) of the Amount column.
- *Why:* To see which specific items bring in the most cash.

The screenshot shows the Power Query Editor interface. The main area displays a table with 17 rows and 2 columns. The first column is 'Sub-Category' and the second column is 'Amount of Subcategory'. The data is grouped by Sub-Category, with each group having a 'List.Sum([Amount])' value. The 'Query Settings' pane on the right shows the 'APPLIED STEPS' list, which includes 'Grouped Rows'.

Sub-Category	Amount of Subcategory
Bookcases	56861
Stole	18546
Hankierchief	14608
Electronic Games	39168
Phones	46119
Saree	53511
Trousers	30039
Chairs	34222
Kurti	3361
T-Shirt	7382
Shirt	7555
Leggings	2106
Tables	22614
Printers	58252
Accessories	21728
Furnishings	13484
Skirt	1946

Finding Monthly Sales Targets

- *What I did:* I copied the Sales Target table and grouped it by the month to get the **Sum** of the Target column.
- *Why:* To calculate the total sales goals for each month.

The screenshot shows the Power Query Editor interface. The main area displays a table with 12 rows and 2 columns. The first column is 'Month of Order Date' and the second column is 'Monthly Sales Target'. The data is grouped by Month of Order Date, with each group having a 'List.Sum([Target])' value. The 'Query Settings' pane on the right shows the 'APPLIED STEPS' list, which includes 'Grouped Rows'.

Month of Order Date	Monthly Sales Target
18-04-2026	31400
18-05-2026	31500
18-06-2026	31600
18-07-2026	33800
18-08-2026	33900
18-09-2026	34000
18-10-2026	36100
18-11-2026	36300
18-12-2026	36400
19-01-2026	43500
19-02-2026	43600
19-03-2026	43800

Connecting the Tables (Data Modeling)

Relationship Lines

I went to the **Model View** canvas and drew lines to connect my tables so they can talk to each other.

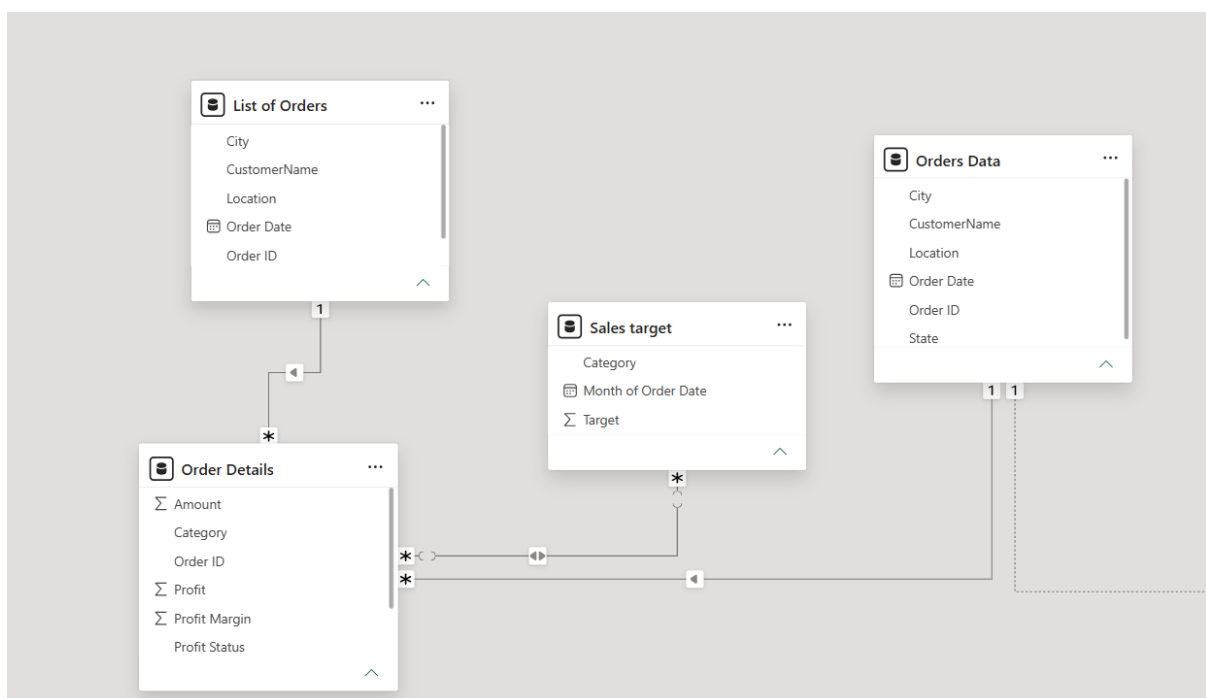
Data Analytics — Power BI Project - 1

- **Connection 1:** List of Orders (Order ID) connected to Order Details (Order ID).
- **Connection 2:** Order Details (Category) connected to Sales Target (Category).

✓	Order Details (Category)	* —> < — *	Sales target (Category)	Active	...
✓	Order Details (Order ID)	* —< 1	List of Orders (Order ID)	Active	...

Testing the Connections

I used the **Manage Relationships** tool to double-check my work. I verified that the lines were active and pointing in the right direction. This ensures there are no calculation errors when using the dashboard.



Conclusion

In this project, I imported raw data, cleaned up messy text, changed columns to the right formats, created formulas for profit, and connected the tables together. Now, the data is perfectly clean and ready to be used to build interactive charts and reports.

Thank You!

Data Analytics — Power BI Project - 1